

Q4 — Graph-Based Multi-Agent RAG

A bounded, inspectable workflow for local Malawi public-health retrieval • September 2026

1. Graph objective and topology

The graph turns the Q3 agent contracts into deterministic control flow. Nodes perform one responsibility and return structured state; conditional edges decide whether to answer, refine retrieval, filter unsafe evidence, or abstain. Local qwen3:8b can implement Verifier and Answerer, while routing, retry limits, citation allow-lists, and injection filtering remain application controlled.

START	Evidence loop	Terminal path
User query	Retriever → Safety → Verifier	Answerer → END
User query	Retriever → Safety → Verifier → refined query → Retriever	Abstain → END after limit/failure
User query	Retriever → Safety removes injected chunks → Verifier	Answerer or bounded retry

Graph state

State field	Purpose	Written by
original_query, active_query, filters	Preserve intent while allowing a focused retrieval rewrite.	START / Verifier route
retrieved_chunks, safe_chunks	Separate ranked candidates from screened evidence.	Retriever / Safety
quarantined_ids	Audit which untrusted chunks were excluded.	Safety
verdict, approved_ids, missing_facts, refined_query	Make evidence decisions and next-query intent explicit.	Verifier
retry_count, max_retries	Guarantee termination; a retry returns to retrieval.	Router / configuration
answer, cited_ids, abstain_reason, trace	Store output and a reviewable execution record.	Answerer / Abstain / all

Invariant. Only safe_chunks can reach verification, and only verifier-approved IDs can reach generation. The original query never changes. Nodes append IDs and decisions rather than sensitive full passages.

2. Nodes and outputs

Node	Operation	Output / invariant
Retriever	Hybrid BM25 + MiniLM retrieval for active_query and filters; retain ranks/scores.	retrieved_chunks and query_used; no answer.
Safety	Scan each candidate as untrusted text; quarantine injection indicators.	safe_chunks and quarantined_ids are disjoint.
Verifier	Check complete coverage; return approved allow-list and focused refinement.	verdict; approved_ids are a subset of safe IDs; missing_facts; refined_query.
Answerer	Generate from approved chunks; public-health tone and exact citations.	Cited answer or deterministic abstention.
Abstain	End safely without generation.	Exact "cannot find in sources" plus reason.

Verifier contract

The Verifier receives the original question and safe evidence. It returns verdict, approved_ids, missing_facts, refined_query, and confidence. "Sufficient" means every requested factual part is supported. Schema validation converts malformed output to "insufficient".

Injection handling

Retrieved instructions are data, never commands. Safety filters flagged chunks before verification. The graph checks the reduced set and may refine once per remaining retry. Repeated unsafe or empty retrieval reaches Abstain. Log IDs, patterns, and routes without sensitive raw content.

Answer validation

The Answerer sees only approved chunks. A deterministic check requires [chunk_id] citations and rejects unknown IDs. Production adds Q3 claim-level faithfulness; a score below 0.90 follows a controlled failure edge, never an open-ended loop.

3. Conditional routing and termination

Condition after Verifier	Next edge	State update
sufficient AND non-empty approved_ids are a subset of safe IDs	Answerer	Pass only approved chunks.
insufficient AND refinement exists AND <code>retry_count < 2</code>	Retriever	Increment counter; set active_query.
unsafe request	Abstain	Set reason; skip Answerer.
No safe evidence and a retry remains	Retriever via Verifier	Refine and increment.
No refinement, malformed verdict, or retries exhausted	Abstain	Exact fallback and END.
Answer citations missing or unapproved	Abstain / END	Discard candidate; record failure.

Retry semantics

Initial retrieval is attempt 0. At most two returns from Verifier to Retriever are allowed, so the graph performs no more than three retrieval attempts. `retry_count` increments before each return edge. Answerer and Abstain are terminal. This prevents latency growth, cycling refinements, and repeated exposure to poisoned passages.

Failure handling and observability

Timeouts, invalid schemas, missing indexes, and empty results become explicit state outcomes. A transient infrastructure retry may occur inside a node under a small timeout budget; it does not reset the graph counter. Log node duration, attempt, retrieved/safe/approved IDs, route, verdict, citations, and terminal reason to nested MLflow runs. Redact raw sensitive text.

Implementation mapping

The companion `snippets_q4.py` provides `GraphState`, node functions, conditional routing, and a dependency-free `run_graph` executor. Its injected retriever maps to the Q2 HybridRetriever; Verifier and Answerer can call local Ollama. The same functions can later be registered in `LangGraph` without changing their state contracts or safety rules.